

TASK 1 – Python File Handling & Automation

Project Title:

Automatic File Organizer using Python

1. Objective

The objective of this project is to automate file organization using Python file handling and exception handling techniques.

2. Technologies Used

- Python
- os module
- shutil module

3. Features

- Reads files from folder
- Creates folders automatically
- Moves files based on extensions
- Deletes temporary files
- Uses try-except error handling

4. Code Explanation

The program reads all files from the current folder, checks their extensions, creates folders automatically, and moves files into categorized folders. Temporary files are deleted. Error handling is implemented using try-except blocks.

5. Source Code

```
#importing os and shutil modules
import os
import shutil

# Get current folder path
path = os.getcwd()

# Dictionary for file types
folders = {
    ".jpg": "Images",
    ".png": "Images",
    ".txt": "Documents",
    ".pdf": "Documents",
    ".mp3": "Audio"
}
```

```
# Read all files
files = os.listdir(path)

for file in files:

    # Skip python file
    if file == "file_organizer.py":
        continue

    # Split file name and extension
    filename, extension = os.path.splitext(file)

    # Check file extension
    if extension in folders:

        folder_name = folders[extension]

        # Create folder if it doesn't exist
        if not os.path.exists(folder_name):
            os.makedirs(folder_name)

        source = os.path.join(path, file)
        destination = os.path.join(path, folder_name, file)

        try:
            shutil.move(source, destination)
            print(f"Moved {file} to {folder_name}")

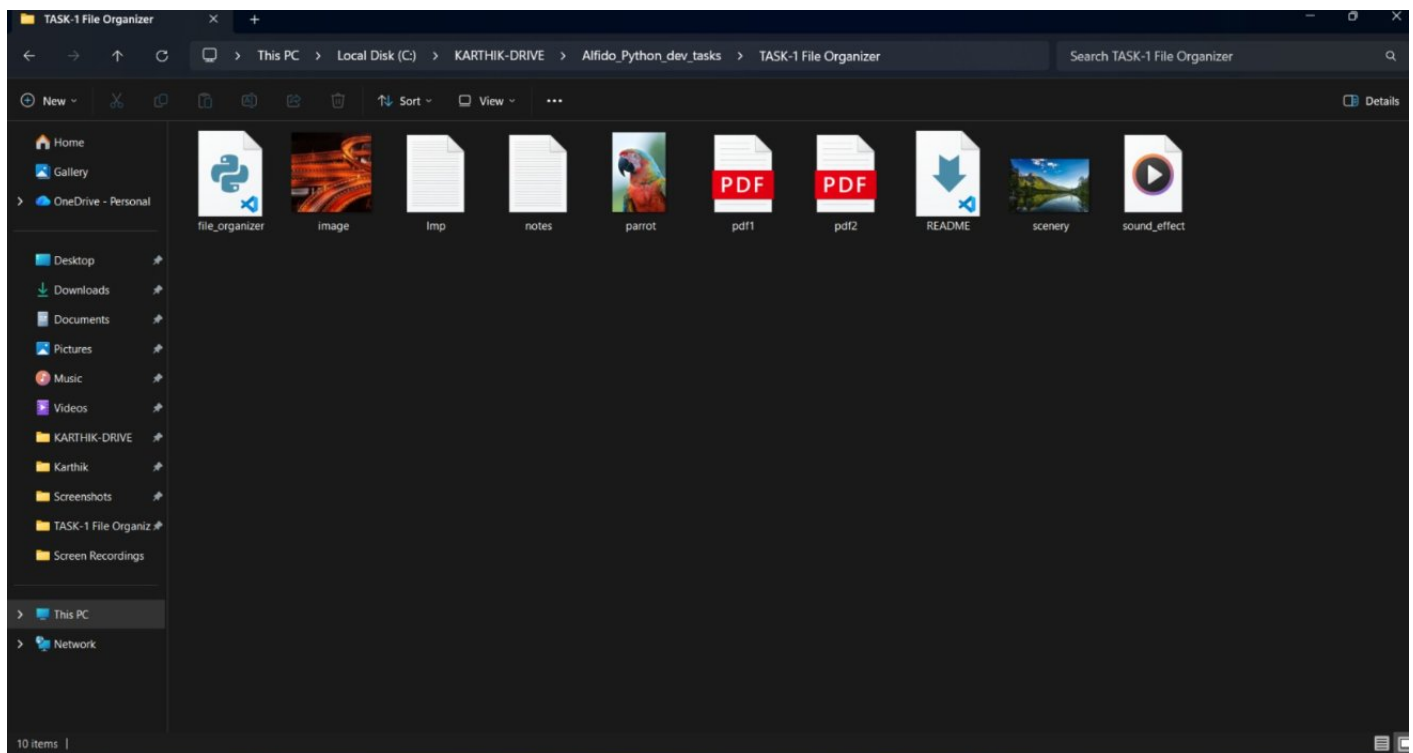
        except Exception as e:
            print("Error:", e)

    # Delete temporary files
    elif extension == ".tmp":

        try:
            os.remove(file)
            print(f"Deleted {file}")

        except Exception as e:
            print("Error deleting file:", e)
```

6. Output Screenshots

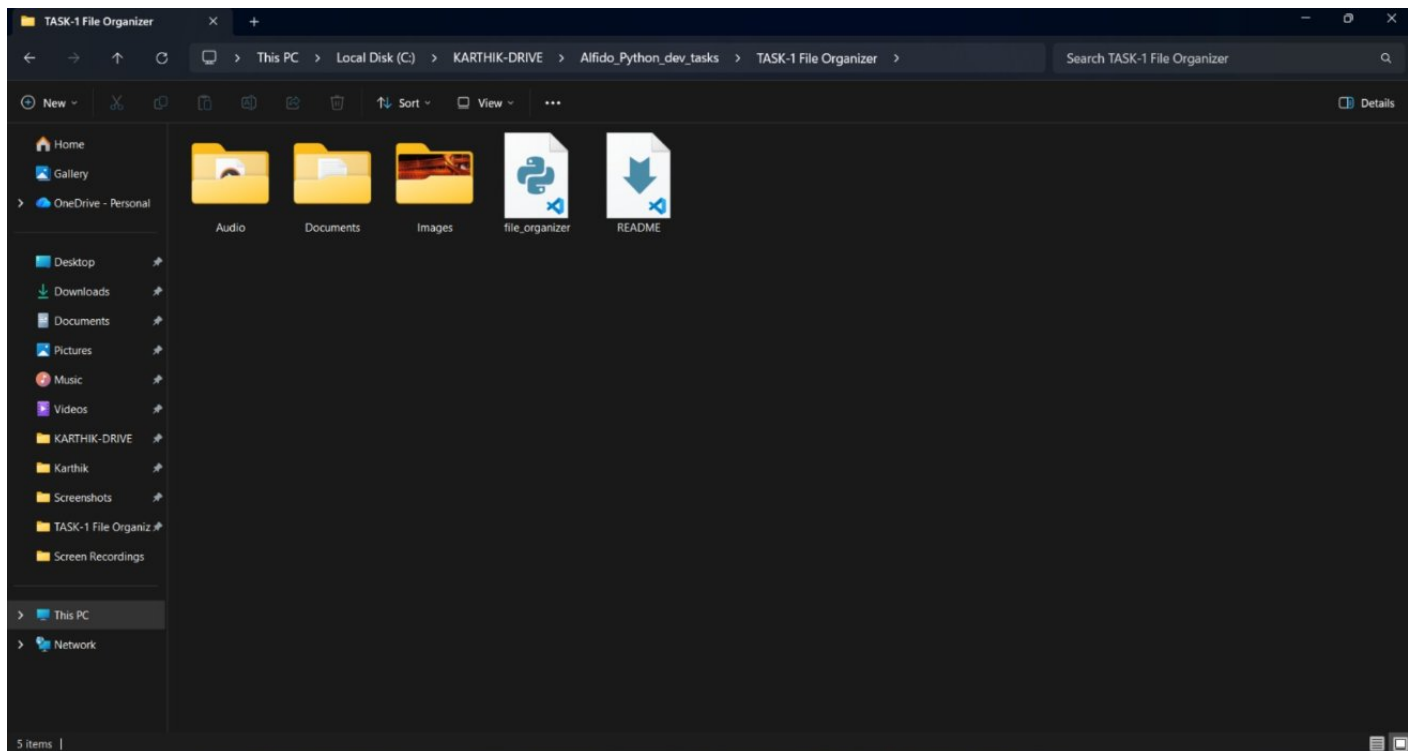


The image shows a Visual Studio Code editor window with the file `file_organizer.py` open. The file is located in the `TASK-1 File Organizer` directory. The code is a Python script that organizes files into folders based on their extension. The terminal output shows the script being executed and the files being moved to their respective folders.

```
1 #importing os and shutil modules
2 import os
3 import shutil
4
5 # Get current folder path
6 path = os.getcwd()
7
8 # Dictionary for file types
9 folders = {
10     ".jpg": "Images",
11     ".png": "Images",
12     ".txt": "Documents",
13     ".pdf": "Documents",
14     ".mp3": "Audio"
15 }
16
17 # Read all files
18 files = os.listdir(path)
19
20 for file in files:
21     # Skip python file
22     if file == "file_organizer.py":
23         continue
24     ...
```

Terminal Output:

```
PS C:\KARTHIK-DRIVE\Alfido_Python_dev_tasks> cd "TASK-1 File Organizer"
PS C:\KARTHIK-DRIVE\Alfido_Python_dev_tasks\TASK-1 File Organizer> python file_organizer.py
Moved image.jpg to Images
Moved Imp.txt to Documents
Moved notes.txt to Documents
Moved parrot.jpg to Images
Moved pdf1.pdf to Documents
Moved pdf2.pdf to Documents
Moved scenery.jpg to Images
Moved sound_effect.mp3 to Audio
PS C:\KARTHIK-DRIVE\Alfido_Python_dev_tasks\TASK-1 File Organizer>
```



7. Conclusion

This project helped in understanding Python file handling, automation, folder management, and error handling using try-except blocks.

8. GitHub Repo link:

<https://github.com/B-Karthik13/Alfido-Tech-Internship/tree/main/TASK-1%20File%20Organizer>

9. LinkedIn:

[Python File Handling & Automation](#)